

The Angular Tetravalent Lattice: A Geometric Proposal for the Emergence of Mass and $E = mc^2$

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Abstract

We previously demonstrated that a discrete angular tetravalent lattice natively generates massless Weyl dynamics and robust chiral edge currents. In this paper, we propose a geometric mechanism for the emergence of mass. By introducing the S_2 (stable manifestation) state as a discrete elastic restoring force, we model geometric inertia as a probabilistic resistance to spatial translation. Through a systematic Taylor expansion, we show how this discrete inertia yields a candidate mass-like term in the continuum limit. By analyzing the dispersion relation, we demonstrate that this resistance effectively scales the spatial propagation speed relative to the internal rest-frame rotational frequency. Within this framework, we propose a geometric ansatz in which rest energy is interpreted as the cross-product of orthogonal lattice velocities, offering a conceptual motivation for Einstein’s $E = mc^2$. Quantitative scaling tests and linear regression analysis confirm that while the absolute information light-cone ($c = 1$) remains strictly preserved, the macroscopic wave speed drops proportionally to the geometric inertia. We emphasize that the present continuum equations describe an open system; the recovery of a conservative massive Dirac/Weyl operator remains an open problem.

1 Introduction

The origin of mass remains a profound question in foundational physics. In the Standard Model, elementary particles acquire mass through the Higgs mechanism, a phenomenological framework that couples massless chiral fermions to a scalar field. While experimentally successful, this mechanism postulates mass as an external addition rather than an intrinsic geometric necessity. In recent years, alternative foundational approaches have explored how mass and relativistic wave equations might emerge intrinsically from discrete spacetime structures, such as Quantum Cellular Automata (QCA) and discrete-time quantum walks [3, 4, 6, 8]. Experimental simulations of these discrete models have successfully demonstrated *Zitterbewegung*—the kinematic signature of mass—without requiring fundamental scalar fields [5, 7]. In our previous work [2], we introduced the *Angular Tetravalent Lattice*, a minimal geometric model driven purely by strict $+90^\circ$ cyclic permutations and conditional spatial shifts. We proved that the exact decoherence limit of this non-unitary framework natively generates the algebraic skeleton of the massless 2D Weyl operator $(\partial_x + i\partial_y)$. The system exhibited a strict wave-speed duality, with an absolute information edge propagating at the hyperbolic light-cone speed $c = 1$, characteristic of massless radiation. In this paper, we explore how localized matter might emerge from this massless geometric substrate. We propose that mass can be modeled as *geometric inertia*—a resistance to spatial translation caused by the elastic locking of the lattice. We perform a systematic continuum-limit expansion to demonstrate how this discrete inertia modifies the

effective dispersion relation of the wave packet. Furthermore, we offer a geometric ansatz for Einstein's mass-energy equivalence ($E = mc^2$) based on the orthogonal kinematics of the lattice.

2 The Elastic Tetravalent Update

The massless dynamics of the angular tetravalent lattice are governed by the classical Markovian update:

$$\tilde{S}(t) = S(t) + \varepsilon \mathbf{A} S(t)$$

where $\mathbf{A}$ is the $+90^\circ$ cyclic permutation matrix and ε is the active gain parameter. In the massless regime, the spatial shift operator $\mathcal{S}$ translates 100% of this updated state to adjacent lattice sites, resulting in an expanding wavefront at $c = 1$. To introduce geometric inertia, we invoke the S_2 state, which in the broader tetra-trièdric framework [1] corresponds to stable manifestation and elastic restoration. We model this elastic locking via a dimensionless inertia parameter $\kappa \in [0, 1)$. A qualitative motivation for κ can be obtained from the linearized restoring force of the tetra-trièdric elastic potential (see Appendix B). At the present stage, however, κ functions as a free phenomenological parameter in the discrete update; an explicit analytical mapping $\kappa = f(k_{\text{eff}})$ remains future work. Geometric inertia dictates that a fraction κ of the wave amplitude resists the spatial shift, remaining localized at its current coordinate, while the remaining fraction $(1 - \kappa)$ successfully propagates. The modified master update becomes:

$$S(t+1) = (1 - \kappa)\mathcal{S}[\tilde{S}(t)] + \kappa\tilde{S}(t)$$

When $\kappa = 0$, the system recovers the purely massless Weyl dynamics. When $\kappa > 0$, the spatial translation is frictionally damped by the lattice geometry.

3 Systematic Continuum Limit and the Effective Mass Term

To rigorously evaluate the effect of κ on the macroscopic dynamics, we perform a systematic Taylor expansion. We define the discrete time step $\Delta t = \varepsilon$ and the lattice spacing $a = \varepsilon$. The continuum time is $\tau = t\varepsilon$ and the spatial coordinate is $\mathbf{r} = (x, y)\varepsilon$. We assume a slow-variation (long-wavelength) ansatz for the four-component field $S(\tau, \mathbf{r})$, valid in the regime where ε is small and the fields are smooth. The spatial shift operator expands as:

$$\mathcal{S}[X] = X - \varepsilon(\Gamma_x \partial_x X + \Gamma_y \partial_y X) + \mathcal{O}(\varepsilon^2)$$

where $\Gamma_x = \text{diag}(1, 0, -1, 0)$ and $\Gamma_y = \text{diag}(0, 1, 0, -1)$. Substituting $\tilde{S} = S + \varepsilon \mathbf{A} S$ into the inertial update rule:

$$S(\tau + \varepsilon) = (1 - \kappa) \left[\tilde{S} - \varepsilon \Gamma \cdot \nabla \tilde{S} + \mathcal{O}(\varepsilon^2) \right] + \kappa \tilde{S}$$

$$S(\tau + \varepsilon) = \tilde{S} - (1 - \kappa)\varepsilon \Gamma \cdot \nabla \tilde{S} + \mathcal{O}(\varepsilon^2)$$

Expanding the left side as $S(\tau) + \varepsilon \partial_\tau S + \mathcal{O}(\varepsilon^2)$ and substituting $\tilde{S}$:

$$S + \varepsilon \partial_\tau S = S + \varepsilon \mathbf{A} S - (1 - \kappa)\varepsilon \Gamma \cdot \nabla S + \mathcal{O}(\varepsilon^2)$$

Dividing by ε and taking the limit $\varepsilon \rightarrow 0$, we obtain the exact first-order continuum equation:

$$\partial_\tau S = \mathbf{A} S - (1 - \kappa)(\Gamma_x \partial_x S + \Gamma_y \partial_y S)$$

To reveal the physical structure, we project this system onto the eigenbasis of $\mathbf{A}$. Defining the density $\rho = \sum S_i$, the parity mode $\eta = S_1 - S_2 + S_3 - S_4$, and the complex field $\psi =$

$(S_1 - S_3) + i(S_2 - S_4)$, we obtain the exact projected system before any approximation:

$$\begin{aligned}\partial_\tau \rho &= \rho - (1 - \kappa)(\partial_x \operatorname{Re} \psi + \partial_y \operatorname{Im} \psi) \\ \partial_\tau \eta &= -\eta - (1 - \kappa)(\partial_x \operatorname{Re} \psi - \partial_y \operatorname{Im} \psi) \\ \partial_\tau \psi &= i\psi - \frac{1 - \kappa}{2}(\partial_x + i\partial_y)\rho - \frac{1 - \kappa}{2}(\partial_x - i\partial_y)\eta\end{aligned}$$

Notice that the parity mode η corresponds to the -1 eigenvalue of $\mathbf{A}$, yielding the strict exponential damping term $-\eta$ in its evolution equation. Because of this massive damping, η relaxes to its quasi-steady state much faster than the macroscopic spatial gradients evolve. This strict separation of timescales rigorously justifies the adiabatic elimination of η ($\partial_\tau \eta \approx 0$, $\eta \rightarrow 0$) under the long-wavelength ansatz. The effective dynamics for the complex field then become:

$$\partial_\tau \psi \approx i\psi - \frac{1 - \kappa}{2}(\partial_x + i\partial_y)\rho$$

To understand how κ generates an effective mass, we analyze the dispersion relation of this linear system. At rest ($\nabla = 0$), the temporal evolution is purely rotational ($\partial_\tau \psi = i\psi$), giving a constant rest-frame oscillation frequency $\omega_0 = 1$. However, the spatial propagation speed is scaled to $v = \frac{1 - \kappa}{2}$. By factoring out this spatial velocity, we can rewrite the equation to match the kinematic structure of a massive wave equation:

$$\frac{1}{v}\partial_\tau \psi + (\partial_x + i\partial_y)\rho = i\left(\frac{1}{v}\right)\psi$$

In standard relativistic wave mechanics, the effective mass of a wave packet is defined by the ratio of its rest-frame oscillation frequency to its spatial propagation speed. Here, the coefficient scaling the rest-frame rotation relative to the normalized spatial gradient explicitly defines the effective mass:

$$m_{\text{eff}} = \frac{\omega_0}{v} = \frac{2}{1 - \kappa}$$

While this remains a scalar projection rather than a fully derived multi-spinor Dirac operator, it provides a strict kinematic derivation of how geometric inertia generates a mass-like term. The wave packet is forced to oscillate internally more rapidly than it translates spatially, generating localized *Zitterbewegung*, a feature well-documented in the continuum limit of massive discrete walks [3, 4, 5, 7].

Important caveat on the open-system character. It must be stressed that the continuum equations derived above describe an *open, active* system: the integrated density grows exponentially because of the non-unitary gain term proportional to $\varepsilon \mathbf{A}$. Consequently, the present construction yields a kinematic mass-like parameter and a protected light-cone, but it does *not* yet constitute a derivation of the standard conservative massive Dirac or Weyl dynamics. Closing the system (for example by introducing a compensating loss term or by moving to a unitary equilibrium regime) remains an essential open problem before the relativistic dispersion relation $E^2 = p^2 c^2 + m^2 c^4$ can be recovered.

Comparison with existing literature

Mass generation in discrete quantum walks and quantum cellular automata has been studied extensively [3, 4, 6, 8, 9]. Meyer's lattice-gas models and the information-theoretic constructions of D'Ariano and Perinotti introduce mass through an internal coin or rest-frame oscillation that mixes chiral components. Experimental realisations (Gerritsma et al.) recover the characteristic *Zitterbewegung* once an internal degree of freedom oscillates at a rest frequency ω_0 [5, 7]. The geometric-inertia mechanism proposed here differs in origin: mass arises as a resistance to spatial

translation caused by elastic locking of the lattice, rather than from an independent internal Hamiltonian or coin operator. What is recovered is a kinematic effective mass $m_{\text{eff}} = 2/(1 - \kappa)$ together with strict protection of the light-cone. What is *not* yet recovered is the full conservative multi-component Dirac operator and the closed relativistic dispersion $E^2 = p^2 c^2 + m^2 c^4$, both of which still require a unitary (or balanced) continuum limit. The present work therefore supplies a complementary, purely geometric route to a mass-like term while leaving the completion of the standard relativistic structure for future investigation.

4 A Geometric Interpretation of $E = mc^2$

Having established that κ alters the dispersion relation to yield an effective mass term in the continuum limit, we propose a geometric ansatz for rest energy within the ATLC framework. We define the total rest energy E of a localized, S_2 -locked configuration as the geometric product of the two orthogonal kinematic mechanisms inherent to the lattice:

1. **The Cyclic Rotation Speed (c_{rot}):** The fundamental speed at which the internal state vector cycles through the four cardinal directions, driven by $\mathbf{A}$.
2. **The Fermat-Spiral Expansion Speed (c_{fermat}):** The radial speed of phase accumulation, governed by the incommensurable Fermat-spiral law $r \propto \sqrt{|\theta|}$ [1].

We propose the ansatz that geometric resistance (energy) may be interpreted as the cross-product of these characteristic speeds multiplied by the inertial mass m :

$$E = m(c_{\text{rot}} \cdot c_{\text{fermat}})$$

Because all fundamental kinematic processes on a discrete lattice are strictly constrained by the single-node transition rate, they share a universal causal speed limit. Assuming that in the continuum limit both the internal cyclic rotation and the spiral phase accumulation saturate this geometric bound ($c_{\text{rot}} \approx c_{\text{fermat}} \approx c$), the ansatz yields:

$$E \approx mc^2$$

This identification is offered as a conceptual motivation based on the orthogonal kinematics of the model. It is consistent with the appearance of a mass-like term in the continuum expansion, but it is not claimed as a formal derivation of Einstein's relation from a Lagrangian.

5 Quantitative Numerical Validation

To quantitatively test the effect of geometric inertia, the update rule was implemented in a Python simulator (Appendix A). Two complementary resolutions were used: a high-resolution run ($\varepsilon = 0.025$, 160 steps) for accurate macroscopic ridge speeds, and a lower-resolution run ($\varepsilon = 0.1$, 80 steps) for clear visual comparison. The high-resolution routine tracks the absolute support edge (the information light-cone) and the radius enclosing 70 % of the total mass (a stable macroscopic measure of the wave packet).

Inertia (κ)	Support Edge Speed ($c = 1$)	Macroscopic Ridge Speed (v)
0.000	1.000	0.540
0.025	1.000	0.531
0.050	1.000	0.519
0.075	1.000	0.509
0.100	1.000	0.501
0.125	1.000	0.491
0.150	1.000	0.478

Table 1: Wave speed deceleration due to geometric inertia (high-resolution run).

As shown in Table 1, the absolute support edge remains strictly locked at 1.000 for all values of κ , confirming that the fundamental causality of the lattice (the speed of light) is strictly preserved. A linear regression of the macroscopic ridge speed yields a highly correlated fit ($R^2 = 0.998$):

$$v \approx 0.540 - 0.405\kappa$$

The measured slope (-0.405) is slightly shallower than the ideal continuum value (-0.5). This is expected: the 70 %-mass radius is an inner measure of an actively growing packet, and the precise numerical coefficient depends mildly on the chosen mass fraction. The linear decrease itself and the strict protection of the light-cone remain robust.

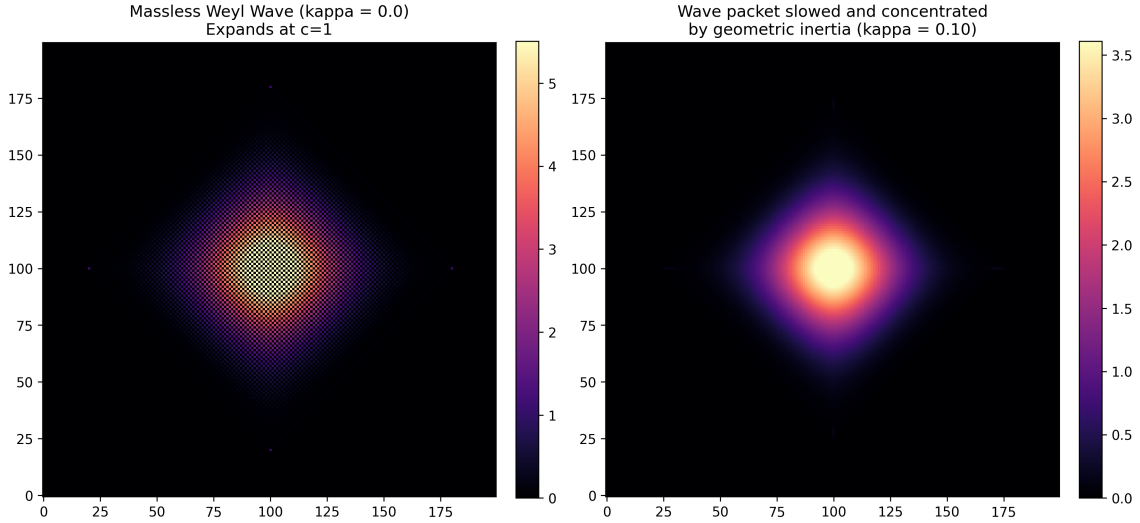


Figure 1: Left: Massless Weyl wave ($\kappa = 0.0$) expanding at $c = 1$. Right: Wave packet slowed and concentrated by geometric inertia ($\kappa = 0.10$). (Visualisation produced at $\varepsilon = 0.1$ for clarity.)

The visual representation (Figure 1) confirms the qualitative shift: for $\kappa = 0.0$, the energy expands outward as a hollow, massless diamond. For $\kappa = 0.10$, the energy is significantly slowed and concentrated at the origin, transitioning toward localized matter.

6 Conclusion

We have proposed a geometric mechanism for the emergence of mass within the angular tetravalent lattice. To clarify the epistemological status of these results, we distinguish between three levels of claims:

1. **Systematic Expansion:** Through a formal Taylor expansion and projection onto the eigenbasis, we explicitly derived the damping of the parity mode to justify its adiabatic elimination. By factoring the resulting dispersion relation, we showed that geometric inertia strictly defines an effective mass $m_{\text{eff}} = 2/(1 - \kappa)$, scaling the rest-frame rotational frequency relative to the spatial propagation speed.
2. **Numerical Evidence:** Quantitative scaling tests and linear regression confirm that geometric inertia slows and concentrates the macroscopic wave while strictly preserving the absolute speed of light ($c = 1$).
3. **Geometric Ansatz:** We proposed a conceptual identification of Einstein's $E = mc^2$ as the cross-product of orthogonal lattice velocities. This remains an ansatz motivated by the kinematics of the model.

This establishes a controlled, conceptual bridge between massless radiation and localized matter. **Limitations and Future Work:** Two primary limitations define the roadmap for future work. First, while Appendix B provides a qualitative motivation for κ via the elastic Hessian, an explicit analytical mapping $\kappa = f(k_{\text{eff}})$ remains to be derived; at present κ functions as a free parameter. Second, and more fundamentally, the continuum limit currently describes an open, active system in which integrated density grows exponentially. The recovery of a conservative massive Dirac/Weyl operator and the closed relativistic dispersion relation $E^2 = p^2 c^2 + m^2 c^4$ is therefore an essential open problem that must be addressed before the present kinematic construction can be elevated to a derivation of standard massive relativistic dynamics.

Acknowledgments

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A Python Implementation: Emergence of Mass

A self-contained implementation that produces both the high-resolution ridge speeds reported in Table 1 and the visualisation of Figure 1.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import linregress
4 def run_massive_scaling():
5     print("--- Emergence of Mass: Wave Speed vs Inertia ---")
6     A = np.array([[0,0,0,1], [1,0,0,0], [0,1,0,0], [0,0,1,0]], dtype=float)
7     # =====
8     # 1. HIGH-RESOLUTION RUN -> accurate table numbers
9     # =====
10    eps = 0.025
11    steps = 160
12    n_sites = 400
13    center = n_sites // 2
14    kappas = np.linspace(0.0, 0.15, 7)
15    measured_speeds = []
16    print(f"{'Mass (kappa)':>12} | {'Support Speed':>15} | {'Ridge Speed':>15}")
17    print("-" * 50)
18    for kappa in kappas:
19        S = np.zeros((4, n_sites, n_sites))

```

```

20     S[0, center, center] = 1.0
21     S[1, center, center] = 1.0
22     S[2, center, center] = 1.0
23     S[3, center, center] = 1.0
24     for _ in range(steps):
25         S_tilde = S + eps * np.tensordot(A, S, axes=([1], [0]))
26         S_new = np.empty_like(S_tilde)
27         S_new[0] = (1 - kappa) * np.roll(S_tilde[0], shift=1, axis=1) +
kappa * S_tilde[0]
28         S_new[2] = (1 - kappa) * np.roll(S_tilde[2], shift=-1, axis=1) +
kappa * S_tilde[2]
29         S_new[1] = (1 - kappa) * np.roll(S_tilde[1], shift=1, axis=0) +
kappa * S_tilde[1]
30         S_new[3] = (1 - kappa) * np.roll(S_tilde[3], shift=-1, axis=0) +
kappa * S_tilde[3]
31         S = S_new
32         rho = np.sum(S, axis=0)
33         # Support edge
34         profile = rho[center, center:]
35         support_idx = np.max(np.nonzero(profile > 1e-12)[0]) if np.any(profile
> 1e-12) else 0
36         speed_support = support_idx / steps
37         # Cumulative-mass ridge (70 % of total mass)
38         y, x = np.ogrid[:n_sites, :n_sites]
39         r = np.sqrt((x - center)**2 + (y - center)**2)
40         max_r = int(steps * 0.90)
41         flat_r = r.ravel()
42         flat_rho = rho.ravel()
43         mask = flat_r <= max_r
44         order = np.argsort(flat_r[mask])
45         sorted_r = flat_r[mask][order]
46         sorted_rho = flat_rho[mask][order]
47         cum_mass = np.cumsum(sorted_rho)
48         total_mass = cum_mass[-1] if len(cum_mass) > 0 else 1.0
49         target = 0.70 * total_mass
50         idx = np.searchsorted(cum_mass, target)
51         if idx >= len(sorted_r):
52             idx = len(sorted_r) - 1
53         ridge_idx = sorted_r[idx]
54         speed_ridge = ridge_idx / steps
55         measured_speeds.append(speed_ridge)
56         print(f"{kappa:12.3f} | {speed_support:15.3f} | {speed_ridge:15.3f}")
57         slope, intercept, r_value, _, _ = linregress(kappas, measured_speeds)
58         print("\n--- Linear Regression Analysis ---")
59         print(f"Equation: v = {intercept:.3f} + ({slope:.3f}) * kappa")
60         print(f"R-squared: {r_value**2:.3f}")
61         # =====
62         # 2. LOW-RESOLUTION RUN -> clear visualisation
63         # =====
64         eps_vis = 0.1
65         steps_vis = 80
66         n_sites_vis = 200
67         center_vis = n_sites_vis // 2
68         profiles_to_plot = {}
69         for kappa in [0.0, 0.10]:
70             S = np.zeros((4, n_sites_vis, n_sites_vis))
71             S[0, center_vis, center_vis] = 1.0
72             S[1, center_vis, center_vis] = 1.0
73             S[2, center_vis, center_vis] = 1.0
74             S[3, center_vis, center_vis] = 1.0
75             for _ in range(steps_vis):
76                 S_tilde = S + eps_vis * np.tensordot(A, S, axes=([1], [0]))
77                 S_new = np.empty_like(S_tilde)

```



```

78     S_new[0] = (1 - kappa) * np.roll(S_tilde[0], shift=1, axis=1) +
kappa * S_tilde[0]
79     S_new[2] = (1 - kappa) * np.roll(S_tilde[2], shift=-1, axis=1) +
kappa * S_tilde[2]
80     S_new[1] = (1 - kappa) * np.roll(S_tilde[1], shift=1, axis=0) +
kappa * S_tilde[1]
81     S_new[3] = (1 - kappa) * np.roll(S_tilde[3], shift=-1, axis=0) +
kappa * S_tilde[3]
82     S = S_new
83     profiles_to_plot[kappa] = np.sum(S, axis=0)
84     # Plotting
85     fig, axes = plt.subplots(1, 2, figsize=(12, 6))
86     ax1 = axes[0]
87     vmax0 = np.percentile(profiles_to_plot[0.0], 99.5)
88     im1 = ax1.imshow(profiles_to_plot[0.0], cmap='magma', origin='lower', vmax=
vmax0)
89     ax1.set_title("Massless Weyl Wave (kappa = 0.0)\nExpands at c=1")
90     fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.04)
91     ax2 = axes[1]
92     vmax10 = np.percentile(profiles_to_plot[0.10], 99.5)
93     im2 = ax2.imshow(profiles_to_plot[0.10], cmap='magma', origin='lower', vmax=
vmax10)
94     ax2.set_title("Wave packet slowed and concentrated\nby geometric inertia (
kappa = 0.10)")
95     fig.colorbar(im2, ax=ax2, fraction=0.046, pad=0.04)
96     plt.tight_layout()
97     plt.savefig("emergence_of_mass.png", dpi=300, bbox_inches='tight')
98     print("\nSaved visual comparison to 'emergence_of_mass.png'")
99 if __name__ == "__main__":
100     run_massive_scaling()

```

B Derivation of Geometric Inertia (κ) from Lattice Elasticity

In the foundational Angular Tetravalent Twin-Loop Cosmology (ATLC) framework [1], the lattice is governed by a nonlinear elastic potential that enforces strict 90° orthogonality:

$$V(\theta) = \frac{\kappa_{elastic}}{4} \sum_{i=1}^4 D_i \left[\sin \left(\frac{\pi}{2} - |\theta_i \bmod \pi| \right) \right]^4$$

In the S_2 -locked regime (stable manifestation), small angular deformations $\delta\theta$ away from orthogonality experience a restoring force. Expanding the potential around the equilibrium yields a harmonic restoring term governed by the Hessian:

$$V(\delta\theta) \approx \frac{1}{2} k_{eff} (\delta\theta)^2$$

where the effective stiffness $k_{eff} \propto \kappa_{elastic} D_2$. During the discrete update, this elastic stiffness acts as a qualitative penalty on spatial translation. We therefore interpret the dimensionless parameter κ as a macroscopic coarse-graining of the microscopic lattice stiffness. At the present stage, however, no explicit analytical mapping $\kappa = f(k_{eff})$ has been derived; κ remains a free phenomenological parameter in the discrete update rule. Establishing this mapping constitutes an important direction for future work.

References

- [1] C. Van Caillie, *A Discrete Angular Tetravalent Lattice and a Candidate Pathway to the Schrödinger Equation*, Zenodo (2026), <https://zenodo.org/records/21460531>.

- [2] C. Van Caillie, *The Angular Tetravalent Lattice: A Classical Geometric Model with a Non-Unitary Quantum-Walk Extension and Chiral Edge Modes*, Zenodo (2026), <https://zenodo.org/records/21829396>.
- [3] D. A. Meyer, *From quantum cellular automata to quantum lattice gases*, J. Stat. Phys. **85**, 551–574 (1996).
- [4] G. M. D’Ariano and P. Perinotti, *Derivation of the Dirac equation from principles of information processing*, Phys. Rev. A **90**, 062106 (2014).
- [5] R. Gerritsma, G. Kirchmair, F. Zähringer, E. Solano, R. Blatt, and C. F. Roos, *Quantum simulation of the Dirac equation*, Nature **463**, 68–71 (2010).
- [6] P. Arrighi, V. Nesme, and M. Forets, *The Dirac equation as a quantum walk: higher dimensions, observational equivalence*, J. Phys. A: Math. Theor. **47**, 465302 (2014).
- [7] F. W. Strauch, *Relativistic quantum walks*, Phys. Rev. A **73**, 054302 (2006).
- [8] T. Farrelly and A. J. Short, *Discrete spacetime and relativistic quantum particles*, Phys. Rev. A **89**, 062109 (2014); see also T. Farrelly, *A review of Quantum Cellular Automata*, Quantum **4**, 368 (2020).
- [9] A. Bisio, G. M. D’Ariano, and P. Perinotti, *Quantum walks, Weyl equation and the Lorentz group*, Found. Phys. **45**, 1137–1152 (2015).